Scenario audit of Cuba's 2021–2026 population decline

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Scenario ensembles, an infant-mortality consistency check, an excess-mortality decomposition, and triangulation of independent registers

Supplementary figures (life expectancy; provinces):
cuba-depopulation-2021-2026-supplement.pdf

Abstract

Background. Cuba is losing population at one of the fastest peacetime rates in the Western Hemisphere. No census has been completed since 2012, and the national statistics office (ONEI) revised its published headcount downward by more than one million in a single 2024 revision. End-2025 published levels still span roughly 8.0–11.0 million. This paper audits the 2021–2025 headcount decline with transparent scenario ensembles, decomposes migration versus internal demographic change, and reports an end-2026 scenario continuation.

Methods. We solve the demographic accounting identity under three scenarios—floor, central, upper—each specified by three numbers: end-2021 roll overstatement, a multiplier on ONEI net emigration, and residual death under-registration. Point figures are closed form and exact by linearity; a reduced Monte Carlo reports prior sensitivity ranges only. No parameter is informed by a likelihood, so these are scenario values, not estimates, and the ranges are not confidence intervals. All results are stated against a single base, the official end-2021 stock. External registers are used as ceilings or soft corroboration; the central scenario is never treated as external validation of itself. Separately, we decompose the change in annual deaths since 2019 by indirect age standardisation on ONEI age-specific mortality, splitting it into a population-size term, an ageing term, and a residual rate term reported as excess deaths relative to the 2019 age-specific schedule.

Results. Against the official end-2021 stock of 11.11 million, the gap by end-2025 spans **1.97–2.79 million** (**17.8–25.1%**) across three assumption sets; a null scenario taking ONEI at face value gives 1.68 million (exactly, by construction) and is reported with them. Two quantities must be kept apart. Against the *official 2021 stock*, the central scenario's gap is **2.53 million** (22.8%)—but 1.68 million of that is decline ONEI itself published, so only **0.85 million** is this paper's added correction. Against ONEI's *end-2025* figure of 9.43 million, the central scenario's living population of **8.58 million** is lower by **0.85 million**. That 0.85 million is the paper's actual claim; the 2.53 million is the two combined. A null scenario (U_0=0, M=1, D_{fac}=1) reproduces ONEI's published figure exactly—necessarily so, since R is defined as the residual that closes the identity—and is reported alongside the three assumption sets so the span does not exclude the official series by construction. About **92%** of that gap is emigration, counting both the 2022–2025 flow and the pre-2021 backlog still on the roll. The end-2026 continuation places the central scenario near **8.29 million**. Cuba registered 27,100 more deaths in 2025 than in 2019, or some 31,300 once estimated unregistered deaths are added, while the central scenario's stock is about 2 million smaller. Age standardisation separates two upward forces of comparable size—about **22,400** additional deaths per year from an older population and about **28,500** per year of excess against the 2019 schedule (prior sensitivity range 18,600–38,200)—against roughly 19,500 fewer deaths per year from the smaller population. Excluding the 2021 pandemic wave, the excess total over 2022–2025 is ∼**74,900**

(47,000–103,100), conditional on a flat 2019 anchor; the choice of counterfactual moves it across 45,200–103,300, an uncertainty of comparable size, orthogonal to that range. No single term inside the simulation dominates either: freezing each in turn, the cohort projection accounts for 28% of the 2025 band width and every other term for 8.4% or less. That scope matters—the single largest lever on the result, the 8% elderly-emigrant share, is held fixed rather than propagated, and on its own it swings the headline by about a quarter of the range.

Conclusions. Depopulation is predominantly migratory in the headcount, reinforced by collapsing births, elevated mortality, and accelerated ageing. Against the official 2021 stock the end-2026 gap approaches one in four; because a single base is used throughout, this is the same quantity as the headline, not a second definition. Mortality is a small share of headcount loss but a large absolute toll: it is separately measurable, roughly half of it is not explained by ageing, and it is still rising.

Keywords: Cuba; demography; emigration; natural decrease; infant mortality; fertility; ageing; statistical data quality.

1 Introduction

Counting Cubans has become contested. The last completed census was in 2012; the census planned for 2022 has been repeatedly postponed and is now promised for 2026 [ONEI, 2022]. Without a census, published population depends on projections and a lagging migration register [Preston et al., 2001, United Nations, Department of Economic and Social Affairs, Population Division, 2024]. In July 2024, ONEI reported an "effective population" of 10,055,968 at end-2023 (about one million below prior published levels), then 9,748,007 (end-2024) and 9,434,593 (end-2025) [ONEI, 2024, 2025].

Three incompatible series coexist (Figure 5a): the United Nations World Population Prospects path near 10.9 million (still assuming net emigration of order 2×10^4 per year) [United Nations, Department of Economic and Social Affairs, Population Division, 2024, Economic Commission for Latin America and the Caribbean (ECLAC), 2023]; ONEI's revised official series (9.43 million at end-2025); and independent studies placing the living population nearer 8.0–8.6 million [Albizu-Campos Espiñeira, 2025, 2024]. Almost three million people separate the extremes. This paper presents a scenario audit of the 2021–2025 headcount decline with transparent prior ensembles, decomposes migratory and internal components, compares official migration accounting with destination registers, and reports an end-2026 scenario continuation.

Cuba's fertility transition was already among Latin America's earliest and deepest [Díaz-Briquets and Pérez, 1982, Hollerbach et al., 1984, Díaz-Briquets, 2014], so below-replacement fertility is not new. What is new is the *speed* of headcount loss under combined emigration and natural decrease, in a regional context where Latin America and the Caribbean still grow overall [Economic Commission for Latin America and the Caribbean (ECLAC), 2023, United Nations, Department of Economic and Social Affairs, Population Division, 2024]. Comparable peacetime contractions elsewhere have been driven primarily by emigration; Venezuela is the nearest regional parallel. We use it only as qualitative context: the magnitudes in Table 5 are not on a like-for-like basis with ours, and no Venezuelan figure enters any calculation in this paper.

Central thesis. The headcount loss is mostly emigration, but a second "internal" engine (birth decline, excess mortality, and ageing) would sustain contraction even if emigration slowed [Lee and Mason, 2014, Lesthaeghe, 2010]. The two engines are not independent: selective exit of working-age adults and women of reproductive age causes much of the fertility decline, so many "births that do not occur" are a delayed migratory effect [Bongaarts and Sobotka, 2012, Díaz-Briquets, 2014].

2 Methods

The balancing equation is the standard demographic accounting identity [Preston et al., 2001]. Registered ONEI aggregates over 2022–2025 are fixed anchors [ONEI, 2025]: births $B^{\text{reg}} = 325{,}233$, deaths $D^{\text{reg}} = 502{,}149$, and identity-consistent net emigration $R = 1{,}501{,}706$. Their provenance is uneven: 2019–2022 vital counts are in the ONEI tables we ship, 2023 deaths are independently corroborated by Cuba's WHO submission, but the 2024 births and deaths (71,374 and 128,098) appear in no table available to us and rest on state reporting. The 2024 death total feeds the 2024 excess figure and every cumulative window, and is the weakest link in the series. Official end-2021 population is $P^{\text{off}}_{2021} = 11{,}113{,}215$. Each Monte Carlo draw samples latent corrections and applies

$$P_{2021} = P^{\text{off}}_{2021} - U_0, \tag{1}$$

$$D = D^{\text{reg}}\, D_{\text{fac}} \;\; (+ \text{ optional epidemic pulse in B/C}), \tag{2}$$

$$M_{\text{net}} = R\,M, \tag{3}$$

$$\Delta = (D - B) + M_{\text{net}}, \tag{4}$$

$$P_{2025} = P_{2021} - \Delta, \tag{5}$$

where U_0 is end-2021 baseline overstatement (people already effectively abroad but still on the roll), $M \geq 1$ scales ONEI net emigration, and $D_{\text{fac}} \geq 1$ scales registered deaths. Loss is reported on a single base: the *official* end-2021 stock. The gap it defines has three additive parts—baseline overstatement U_0, net emigration $R\,M$, and natural decrease—so no second "corrected base" percentage is needed.

Each scenario is three numbers, and the point figure is the arithmetic above evaluated at their prior means, which is exact because the identity is linear in all three. We verified this: the closed form and a Monte Carlo over the same margins agree to within about a thousand people (1,117 in the upper scenario), seed noise with mean zero rather than a systematic gap—across seeds the difference changes sign. The copula has therefore been removed—it moved the median by ∼2,200 people and only widened a within-scenario band that is not our primary uncertainty statement. A reduced draw ($N = 2 \times 10^5$, independent margins) is retained solely to report that band, which remains a *prior-predictive* spread under fixed Beta margins, not sampling error. The **across-scenario envelope** is the primary uncertainty statement [Pérez-Riverol, 2026d,b].

We define three scenarios. **Conservative** adds a modest correction. It is *not* an ONEI-consistent lower bound: its mean M is 1.089, i.e. 9% more emigration than ONEI reports. The zero-correction case is the null scenario. **Central** raises the migration prior above ONEI and caps residual death under-registration at 9%—just above the 7.5% the register itself shed in 2022 without explanation. **Upper** defines the top of the envelope. Two other prior sets have been retired: a crisis-adjusted variant landed within 0.02 million of the central scenario—the same scenario with different dials—and a vital-reconstruction companion built on a hand-weighted deterioration index has been superseded by the attributable-mortality decomposition below, which estimates the same quantity from observed age-specific data rather than from a dimensionless score. These are **scenarios** (prior narratives), not identified estimators of a unique living population. ONEI's 2023–2024 revisions already absorbed a large lagged stock, so scaling revised R by $M > 1$ *and* subtracting a large U_0 can double-count the same outdated-register defect; we therefore always report the floor beside the central path and treat $U_0 + (M - 1)R$ as constrained by destination floors plus register lag.

Infant mortality rose from 5.0 to 9.9 per thousand live births (2019–2025), a ∼98% increase in the *rate*. The 2025 value is a reported figure and is not in any ONEI table available to us. Most of that is the denominator: infant *deaths* rose from 552 (observed, ONEI 3.16) to ∼674 (derived), about 22%, while live births fell 38%. Both figures matter and only reporting the rate overstates the signal. Infant deaths are comparatively hard to hide (hospital occurrence; international scrutiny), so the infant rate is often treated as a

Table 1: Central-scenario specification. Three margins, no dependence structure. Beta scales: $U_0 \in [0, 700{,}000]$, $M \in [1, 1.72]$, $D_{\text{fac}} \in [1, 1.09]$. Birth noise independent $B = B^{\text{reg}}\mathcal{N}(1, 0.01^2)$. The point figure uses the prior means of these three margins.

Symbol	Margin	Interpretation
U_0	$700{,}000 \times \text{Beta}(2.0, 2.4)$	End-2021 roll overstatement
M	$1 + 0.72 \times \text{Beta}(2.2, 2.4)$	Multiplier on ONEI net emigration R
D_{fac}	$1 + 0.09 \times \text{Beta}(1.8, 3.2)$	Residual death under-registration (at-home elderly channel)

high-integrity sentinel of health-system *stress*. Two caveats temper that: the 2025 value is press-reported, as noted above; and there is a literature questioning late-fetal/early-neonatal classification at the viability margin in Cuba [Berdine et al., 2018], so part of any rise may be relaxation of classification practice rather than deterioration. Over the same period the registered crude death rate rose from 9.7 to 14.2 per thousand (∼46%). Both moved sharply upward together, which is what a deteriorating system with an intact death register looks like; a register concealing deaths at scale would show the sentinel rising while all-cause mortality stayed flat. A log-elasticity calibrated on Venezuela, carried in earlier versions, has been removed: it rested on two numbers from one country with no uncertainty attached, and the conclusion—no signature of large-scale concealment—follows from the co-movement without it. Residual under-registration in the central scenario is accordingly limited to a small at-home elderly channel (central ∼3%, cap 9%) [Pérez-Riverol, 2026b].

Mortality is a separate estimand from the headcount, answering a separate question: how much of the rise in deaths is explained by the resident population being older, and how much is not? A crude-rate baseline cannot answer this, because it charges the entire ageing shift—from 20.4% aged 60+ in 2019 to roughly 25–27% in 2025—to the crisis. We therefore replace the crude bridge used in earlier versions of this work with indirect age standardisation [Preston et al., 2001].

Let m_a^{2019} be age-specific death rates in 2019 from ONEI table 3.15, over the groups $a \in \{0\text{–}14,\ 15\text{–}59,\ 60\text{–}64,\ 65+\}$—the finest set the deaths table shares with ONEI table 3.3, which publishes mid-year population by age annually. Splitting 60+ matters: the 1960s birth cohort is now entering at 60–64, so the elderly group is becoming *younger* (60–64 rose from 25.7% of the 60+ population in 2019 to 27.9% in 2022), and a single open-ended 60+ group would misread that composition shift as rising mortality. Expected deaths in year t under an unchanged 2019 health system are $E_t = \sum_a m_a^{2019} P_{a,t}$, and the change in annual deaths since 2019 decomposes exactly into

$$\text{size} = (P_t - P_{2019})\,\text{CDR}_{2019}/1000, \tag{6}$$

$$\text{ageing} = \textstyle\sum_a m_a^{2019}(P_{a,t} - P_{a,2019})/1000 - \text{size}, \tag{7}$$

$$\text{rate} = D_t\, D_{\text{fac}} - E_t, \tag{8}$$

with $D_t D_{\text{fac}} - E_{\text{ref}} = \text{size} + \text{ageing} + \text{rate}$, where $E_{\text{ref}} = \sum_a \tilde{m}_a P_{a,2019}$ applies the same perturbed schedule to the 2019 structure. Against the *registered* 2019 count the identity does not close per draw—the schedule noise does not cancel, leaving a ±5,000 spread in 2025—which is why the reference is E_{ref} and not D_{2019}. This is a Kitagawa-type decomposition applied to a death *count* rather than to a crude-rate difference; the size term is the count analogue of holding exposure fixed [Kitagawa, 1955]. The rate term is *excess deaths relative to the 2019 age-specific schedule*. We deliberately do not call it "attributable to health-system deterioration". The same formula applied to 2021 returns ∼61,900, which is a pandemic, not a health system; a residual against a fixed schedule absorbs pandemic and post-acute mortality, coding drift, mortality displacement, health-selective emigration, and denominator error without discriminating among them. Attribution would require an identified exposure, cause-of-death evidence consistent with the mechanism, and quantitative exclusion of competitors—none of which we have.

The decomposition is far less denominator-sensitive than the crude bridge it replaces. In 2019, 82% of Cuban deaths occurred at ages 60+, while emigration is $\sim$77% aged 15–59. We therefore hold the elderly *count* approximately fixed across stock scenarios, so that under the central-scenario stock the same elderly people sit on a smaller total and the 60+ share rises to $\sim$29% (against $\sim$26% officially). That $\sim$29% is an assumption of this decomposition, not an observation. The projection does *not* hold the count fixed—it removes elderly emigrants at the 8% share—and doing so *raises* the result: at a 0% share (a genuinely fixed count) the 2025 excess is 23,700 and the 2022–2025 total 62,900, against 28,500 and 74,900 as used, and at 16% they are 33,400 and 86,900. That single expert prior therefore swings the headline by $\pm$4,800 a year and $\pm$12,000 cumulatively—about a quarter and a fifth of the respective prior sensitivity ranges. Consequently E_t moves only 3,070 deaths (2.7%) between the official ($\sim$9.4 M) and central-scenario ($\sim$8.6 M) stocks, against $\sim$7,400 for a crude 2019 rate—so indirect standardisation removes about 60% of the denominator sensitivity, not all of it. That residual 3,070 is 11% of the 2025 headline: less exposed to the migration dispute than the crude bridge, but not insulated from it. The 60–64 and 65+ counts are *observed* for 2019–2022 from ONEI 3.3 and cohort-projected for 2023–2026 from the observed 2022 structure (the 0–14 / 15–59 split is imposed at the fixed 2019 ratio in every year, worth $\leq$16 deaths in 2022), with entrants from the 55–59 cohort, exits at the 2019 schedule, and elderly emigration at the 8% share of emigrants aged 60+ (an expert prior, not a measurement). Two projection choices push the result down and two push it up. Downward: ages 40–59 receive no emigration or mortality depletion and U_0 is not removed from the 2022 age structure, which inflates the projected elderly stock and lowers the excess by roughly 1,000 deaths in 2025. Upward: the 8% elderly share is applied to the 2023 net-migration figure of 1,006,000, which this paper elsewhere argues is a register catch-up, not a flow—removing $\sim$80,000 elderly in one year and raising the 2025 result by about 2,450—and the 35/65 split of elderly emigrants between ages 60–64 and 65+ sits at the end of its plausible range that maximises the excess (a 50/50 split lowers 2025 by $\sim$750). We have not reconciled the 2023 treatment with the register-catch-up argument and flag it as an inconsistency. We never derive the elderly count as (age share $\times$ official total): the official total absorbs ONEI's 2023 register catch-up in a single step, and multiplying that step by an age share deletes elderly residents who never emigrated. We propagate four sources of uncertainty over $N = 2 \times 10^5$ draws: 2% noise on the 2019 schedule; a symmetric $\mathcal{N}(0, 0.03)$ residual drift for within-65+ composition after 2022; the central-scenario D_{fac} prior for unregistered deaths; and a 50/50 mixture over official and central-scenario population paths, with a projection error growing 1.5% per year beyond 2022. That error term is symmetric and therefore cannot represent a bias. Run forward from observed 2019, the rule overshoots the observed 60+ count at every horizon (by 1.4%, 3.4% and 5.1% with no emigration; 0.7%, 2.1% and 3.1% at the 8% elderly share). That overshoot is *not* mainly a defect of the cohort flow: substituting each year's actual age-specific rates for the 2019 schedule collapses the horizon-two error in the 8% arm from $+49{,}399$ to -101, because the overshoot is dominated by unmodelled 2020–2022 excess mortality that the 2019 schedule does not contain—$+$48,120 of it in 2021 alone. A back-test window whose middle year is a pandemic cannot calibrate a 2023–2026 horizon, so we publish no magnitude from it. The mechanism it points to is nonetheless real: depleting the elderly stock at the 2019 schedule under-depletes it whenever actual mortality runs above 2019, which raises E_t and therefore lowers the reported excess. Applying the 8% elderly share to the 2023 catch-up figure pushes the other way by a comparable amount; we claim no net sign for the projection as configured. An earlier version of this paper quoted a death count from this exercise; it is withdrawn. Systematic parameters are drawn once and reused in every year, so cumulative windows propagate correlated rather than independent error. The identity above closes exactly in every draw (maximum residual $< 10^{-10}$) because the E_{ref} reference uses the same perturbed rates as the target year. Because D_{fac} is shared with the central scenario, the excess estimate and the headcount scenario are not independent. One further choice must be stated: D_{fac} lifts the target years but not the 2019 anchor, so it enters as a *deterioration in registration completeness since 2019* rather than as a constant level. That is the reading we intend—a crisis-induced at-home elderly channel that did not exist in 2019 genuinely produces excess deaths—but it raises the result. Applying D_{fac} symmetrically to the anchor gives 24,900 for 2025 and 60,600 for 2022–2025, about 13% and 18% below the headline. A crude registered-excess comparator on a linear-drift counterfactual, also carried in earlier versions, has been dropped: it did not reproduce from the stated formula, its drift was below Cuba's own observed pre-crisis crude drift, and it served no argument the age-standardised decomposition does not serve better [Karlinsky and Kobak, 2021, Msemburi et al., 2023, Kishore et al., 2018].

Published population levels are grouped by information content (Figure 4). Ceilings that do not purge emigrants include the UN World Population Prospects, the MINSAP denominator, and the electoral roll (soft upper bounds only). The official ONEI revised headcount is 9.43 M at end-2025. Exit-correcting or independent constructs include housing$\times$occupancy ($\sim$8.7 M; fragile occupancy scalar) and Albizu-Campos 2023/2024 (8.0–8.6 M; methods that share migration-critique assumptions with this paper). The central scenario (8.58 M) is the study estimand and is *not* external validation of itself; agreement with Albizu/housing is partly shared-prior echo, while ceilings bound the upside. The soft band from exit-correcting sources is near 8.0–8.9 million.

The settled-concept destination floor (Table 2) comprises United States $\sim$800,000 (administrative/CBP-pipeline statuses, *not* ACS stock), non-U.S. total $\sim$250,000 (Spain $\sim$135,000 + Uruguay $\sim$35,000 + rest $\sim$80,000), and floor $\sim$1.05 million [Pérez-Riverol, 2026a]. Pew Research Center [2023] is cited for ACS-based context on the settled Cuban-origin population only; it is a different concept from the administrative statuses above and is not the source of the $\sim$800,000. ACS Cuban-born stock rose by only $\sim$335,000 from 2019–2024—a lower stock-change floor that must not be silently equated with the 800,000 administrative figure. Encounter counts and nationality applications are separate concepts and are never added into the settled floor [U.S. Customs and Border Protection, 2024].

Table 2: Bridge from settled destination floor toward scenario migration engines. No row is ledger-like. The floor is the most evidence-proximate, but it is assembled by the authors from approximate destination totals (US $\sim$800,000 administrative settled status, Spain $\sim$135,000 padrón, Uruguay $\sim$35,000 residences, rest of world $\sim$80,000) rather than taken from a single published table; the constituent figures are rounded and their retrieval dates are recorded in the repository. Later rows are residuals or priors.

Concept	Approx. count	Role
Settled destinations floor (US admin + non-US)	1.05 M	Lower bound (settled concept)
U.S. ACS Cuban-born rise 2019–24 (not settled)	$\sim$0.34 M	Stock-change floor; do not add
Unquantified gap toward ONEI R	$\rightarrow\sim$1.5 M	Residual, not validated persons
ONEI identity-consistent net R (2022–25)	1.50 M	Official flow after revision
Albizu-Campos–type net	$\sim$1.8 M	Independent upper narrative
Central-scenario median $R \times M$	$\sim$2.0 M	Prior-driven scenario engine
U.S. encounters 2022–2024 (not settled)	$\sim$0.85 M	Events; do not add to floor
Spain nationality applications	$\sim$0.30 M	Many applicants still in Cuba

Destination data do *not* identify the central scenario's $\sim$2.0 million net emigration; they support a large floor and leave the 1.05$\rightarrow$2.0 path as soft gap-filling plus a migration prior. If true net emigration were nearer 1.5 than 2.0 million, living population would sit toward $\sim$9.1 rather than $\sim$8.6 million; cumulative loss would remain large and emigration would still dominate absolute loss under every scenario in Table 4. Conditional on end-2025 draws, the end-2026 scenario continuation uses births centered at 65,682 ($\mathcal{N}$ noise

3%), deaths centered at $1.04 \times 136{,}214$ times the same D_{fac}, and a three-regime net emigration mixture with probabilities $(0.35, 0.35, 0.30)$ on centers $(150{,}000,\ 210{,}000,\ 300{,}000)$ plus 15% log-scale noise. The $\sim$4% death lift and regime weights are scenario assumptions, not a validated forecast. Paths to $\sim$7 million by 2030 in Figure 5a are **illustrative trend scenarios** (visually the post-2026 segment); they are not probabilistic forecasts.

3 Results

Table 3: Headline quantities by claim. Bands are 90% prior-predictive under the stated margins, not sampling error. "Scenario" marks a quantity that depends on an assumption rather than an observation.

Claim	Quantity	Value	Status
Migration			
	Settled destination floor, author-assembled	1.05 M	lower bound
	ONEI identity-consistent net, 2022–25	1.50 M	official
	Central-scenario migration engine	$\sim$2.0 M	prior
	Emigration share of the 2022–25 flow (central)	$\sim$91%	scenario
	ONEI declared net migration, 2021 / 2022	+169 / +991	official
	US-recorded Cuban encounters, 2021 / 2022 (calendar year)	54,818 / $\sim$313,500	compiled
	High-severity consistency failures	7 of 18	audit
	resting on ONEI's own arithmetic	2 of 18	audit
Mortality against the 2019 age-specific schedule			
	Registered deaths, 2025	136,214	official
	including estimated unregistered ($\times D_{\text{fac}}$)	140,400	scenario value
	Expected at 2019 rates, 2025 age structure	112,100	derived (projected structure)
	Excess vs 2019 schedule, 2025	28,500 [18,600–38,200]	scenario value
	Excess, 2022–2025 (post-pandemic)	74,900 [47,000–103,100]	scenario value
	under alternative counterfactuals	45,200–103,300	sensitivity
	on the pre-revision 2022 total	$\sim$83,800	sensitivity
	Explained by ageing alone, 2025	22,400	derived (projected structure)
	Population-size effect, 2025	−19,500	scenario-dependent
	Standardised mortality ratio vs 2019, 2025	1.215	derived (registered deaths only)
	Share of crude death rate that is age structure (2019)	40.8%	derived
	2022 register removal	8,951 (7.5%)	reported
Projected population			
	Gap vs official 2021, across assumption sets	1.97–2.79 M	scenario span
	of which ONEI's own published decline	1.68 M	official
	this paper's added correction (central)	0.85 M	scenario value
	Living population end-2025 (central)	8.58 M	scenario
	Gap, central scenario	2.53 M (22.8%)	scenario
	ONEI official end-2025	9.43 M	official
	Living population end-2026 (central)	8.29 M	scenario
	Emigration share incl. pre-2021 backlog (central)	$\sim$92%	scenario

3.1 Migration

Closing the gap to ONEI's $\sim$1.5 million is an unquantified residual; the central scenario's $\sim$2.0 million is a prior on R, not destination evidence. If true net is nearer 1.5 than 2.0 million, living population sits toward the high end of our band ($\sim$9.1 rather than 8.6 M).

ONEI's own numbers contradict the destination registers before any model is applied (Figure 1). For 2021 and 2022 ONEI declares net *arrivals* of +169 and +991, while the United States alone recorded 54,818 and roughly 313,500 Cuban *encounters* in those calendar years [U.S. Customs and Border Protection, 2024]. Encounters count events, not unique settled persons, and this paper never adds them into the settled floor; the contradiction survives the distinction, because no fraction of a number that size is reconcilable with a declared net inflow of +991. A systematic pass over the published series (`scripts/consistency_model.py`) returns 18 tests, of which 7 fail at high severity. Only *one* of these rests purely on official sources: T4, the 2023 rate-versus-denominator contradiction, where the rate and the population series cannot both be right. T2 records a 7.5% spread across three 2022 death totals (129,049, 120,098 and 120,108), but that spread is created entirely by the first figure, which is press-reported rather than drawn from an ONEI table; the two official totals differ by ten deaths, or 0.008%. We therefore do not claim T2 as an internal contradiction in ONEI's own published series. Two failures rest on ONEI's own arithmetic: T4, and T3's 2023 seam, where a 1,033,543 single-year population drop sits against a 27,347 natural decrease—both published figures—though grading T3 as severe additionally requires an external bound on plausible annual emigration. The honest count is therefore two internal, one of which needs an outside number to call a failure, not the three the row above might suggest. The 2023 series shows a single-year population drop of 1,033,543 against a natural decrease of 27,347, implying a migration component of 1,006,196—more than double any documented annual outflow, and best read as a backlog correction booked as one year's flow. The published 2023 crude death rate of 11.5 per thousand implies a mid-year population of 10.24 million against 10.57 million in the published series, a gap of 334,000; the two figures cannot both be correct.

Cause-specific standardisation is possible for 2021–2022 only, using Cuba's own ICD-10 submission to WHO over ONEI 3.3 denominators, and we report it because it constrains the reading above more than it supports it. Cuban crude cause-specific rates run about 1.9 times their age-standardised values, so comparisons of Cuban crude rates against regional age-standardised rates—common in public commentary—overstate Cuba's position roughly twofold. Cuba's own standardised rates for 2022 are 87.2 per 100,000 for ischaemic heart disease and 50.4 for cerebrovascular disease. We deliberately do *not* form a ratio against regional comparators: the figures usually quoted (of order 105 and 34) are not reproducible from any source we hold, and the WHO raw files contain no Latin American country with both post-2020 deaths and post-2020 population, so the comparison cannot currently be made on a like-for-like basis in either direction. Between 2021 and 2022 *every* standardised cause fell—all causes by 28%, respiratory by 57%, circulatory by 24%, ischaemic heart by 26%, cerebrovascular by 14%, diabetes by 36% and neoplasms by 6%—because 2021 is a pandemic peak, which is the baseline problem that makes a two-year window uninformative about the 2024–2025 deterioration. We therefore draw no inference from the individual cause movements in either direction. One diagnostic does survive, with a scope limit that matters: ill-defined causes (R00–R99) hold at 0.86–0.98% across 2021–2023 [World Health Organization, 2026], which is evidence against certification deterioration *in those years*. No cause data exist for 2024–2025, where the excess is concentrated, so coding drift in exactly the years that drive the headline remains untested. Cause-specific work cannot extend past 2022 because ONEI 3.3 denominators stop there, which is also why no cause enters the excess estimate.

About **77%** of emigrants are aged 15–59 and the median leaver is about 30 years old (Figure 1); both are expert priors synthesised from the Albizu-Campos reports rather than measurements, as is the 8% elderly share that drives elderly emigration in the projection. Among those remaining, the reported official share aged 60+ is about 26.7% (median age ∼45); ONEI table 3.12 projects 25.0% for 2025, and 26.7% is a reported figure we cannot reproduce from the published tables—neither is a central-scenario-consistent stock. Exit hollows the centre of the pyramid and removes those who would have children [Coleman, 2006, Lee and Mason, 2014]. The flow is majority female [Albizu-Campos Espiñeira, 2025] and concentrated at working ages (Figure 1). Subnational outflow is uneven: ONEI-reported Havana net emigration is about −**33 per 1,000** in 2025 (provincial ranks more reliable than levels if national migration accounting is biased; map in the supplement). Selectivity of exit and the ageing it produces are two views of the same process (Figure 1).

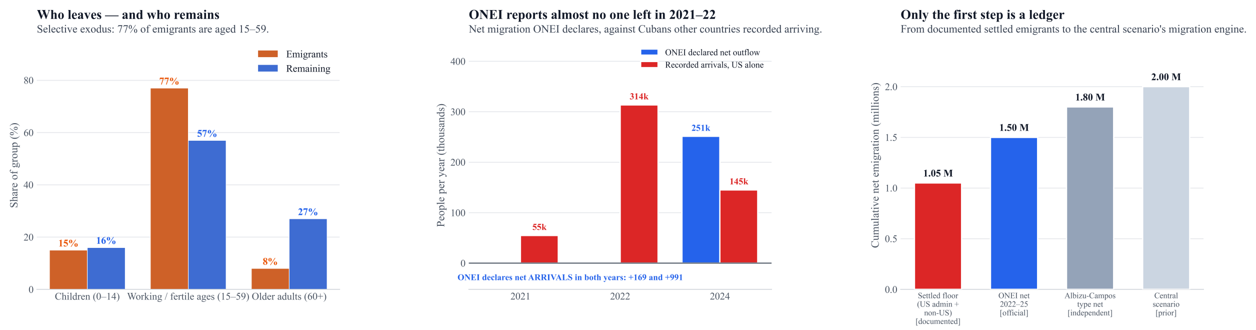
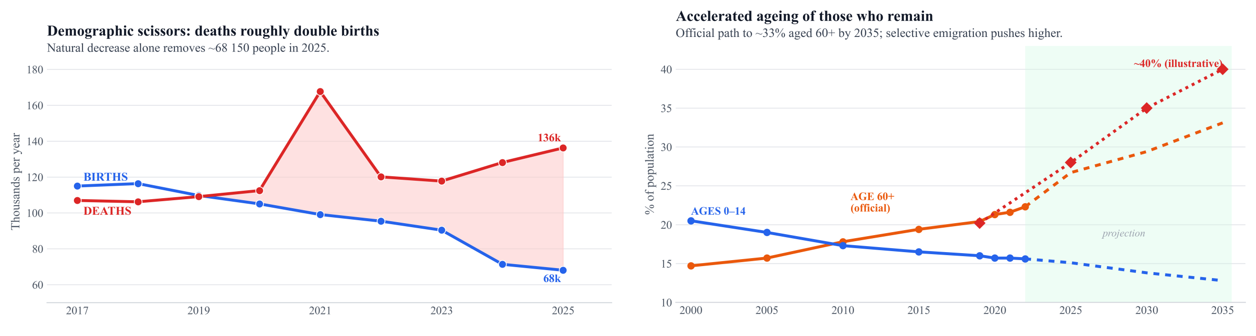


Figure 1: The migration claim, in three panels. Left: age composition of emigrants against remaining residents—the exodus is selective. Centre: net migration ONEI declares against Cubans other countries recorded arriving; ONEI reports net *arrivals* in 2021 and 2022. Right: the bridge from an author-assembled settled floor to the central scenario's migration engine. The first bar is the most evidence-proximate but is not a single published ledger. The working-age hollowing in the left panel is what drives the ageing term in Figure 3.

3.2 Internal demographic engine and excess mortality

Reducing the crisis to emigration alone is incomplete: emigration can reverse, but missing births and an aged residual population cannot. A useful distinction is **realized loss** (exit-dominated) versus **future inertia** (natural decrease increasingly important). In 2019 the natural balance was still slightly positive (+636); in 2025 deaths exceeded births by **68,150** (68,064 births versus 136,214 deaths; Figure 2a). That 2025 deficit matches the 2021 COVID peak without a pandemic: natural decrease is now structural. The total fertility rate fell to 1.29 children per woman in 2025, the lowest since 1958 and well below replacement (2.1) [Díaz-Briquets, 2014, Bongaarts and Sobotka, 2012, Lesthaeghe, 2010]. Emigration removes young women (majority-female flow) and those who remain defer births under scarcity. Births fell 38% (from $\sim$110,000 in 2019 to 68,000 in 2025). A back-of-envelope split into "half fewer women / half lower TFR" is *illustrative only*: ONEI TFR denominators may themselves be inflated by absent women, which would overstate the rate half; a formal exposure–ASFR decomposition is not claimed here. The reported share aged 60+ moves from 20.4% (2019) to 26.7% (2025)—a figure we noted above is not reproducible from the published tables, against ONEI 3.12's own projection of 25.0%—and toward 33% by 2035 under a larger living stock than the central scenario [ONEI, 2025, Lee and Mason, 2014]. Under a smaller, youth-depleted stock the elderly share would be higher; any $\sim$29% / $\sim$40% figures in companion graphics are **illustrative sketches**, not cohort-component estimates. Median age reached **45** years in 2025 on official reporting (Figure 2b).



(a) Births versus deaths ("scissors"): deaths roughly double births by 2025.

(b) Accelerated ageing as working-age adults leave (official structure).

Figure 2: Internal demographic engine: natural decrease and accelerated ageing.

Official life expectancy at birth peaked in 2011–13 (78.5 years) and fell to 77.7 in 2018–20; an independent 2021 estimate places it near 71 years [Albizu-Campos Espiñeira, 2023]. That figure is the lowest of his

2021 variants—he also reports 73.14 excluding COVID-19 deaths, and 73.90 for women—so it should be read as the low end of his own range, not as a consensus. A widely quoted 73.7 for Cuba in 2021 comes from a different source, the UNDP Human Development Report, not from this author. The long-run Cuban series is analysed in Brønnum-Hansen et al. [2023], which covers 1955–2020 and does not extend to 2021. That decline is large relative to COVID-era international losses [Aburto et al., 2022, Msemburi et al., 2023]; LE re-expresses mortality rather than independently corroborating the central scenario (series in the supplement). Infant deaths rose from 461 (2018) to 715 (2022); the rate reached 9.9 per thousand in 2025 ($+98\%$ since 2019).

The mortality decomposition answers the ageing-versus-collapse question directly (Figure 3). Cuba registered 136,214 deaths in 2025 against 109,080 in 2019—27,134 more, or some 31,300 once estimated unregistered deaths are added—while the central scenario's stock is about 2 million smaller (the official series falls by 1.76 million over the same span). The 60+ population meanwhile rose from 2.31 million (2019, observed) to a projected 2.60 million in 2025. Three terms account for that rise (medians are not additive, so they sum to 31,500 rather than 31,300). A resident population smaller by $\sim$2 million should have reduced annual deaths by about **19,500**; ageing added about **22,400**; and the residual excess against the 2019 schedule is about **28,500** per year (prior sensitivity range 18,600–38,200), or $\sim$297 per 100,000 on the official stock. The two upward terms are of comparable magnitude ($\sim$44% and $\sim$56% of their own sum, not of the observed rise); neither alone accounts for the rise, and together they exceed it because the shrinking population pulls the other way. The crude bridge reported in earlier versions of this work charged the whole ageing shift to the crisis and overstated the collapse component: $\sim$77,000 for 2024–2025 combined against **50,100** (32,600–67,500) here.

The time profile separates two distinct episodes. Excess deaths against the 2019 schedule were about 5,400 in 2020, peaked at $\sim$61,900 in the 2021 COVID-19 wave, fell back to $\sim$12,800 (2022) and $\sim$11,900 (2023), and then roughly doubled to $\sim$21,600 (2024) and $\sim$28,500 (2025). The 2022–2023 trough separates the pandemic wave from the later rise, so the 2024–2025 increase is not a continuation of the 2021 peak; it coincides with the blackout, fuel, medicine, and arbovirus crises, though the residual cannot by itself exclude post-acute COVID-19 mortality or coding drift as contributors. Excluding 2021 entirely, the 2022–2025 excess total is $\sim$**74,900** (47,000–103,100); including it, 2020–2025 gives $\sim$142,100 (105,400–180,500). A 2026 scenario continuation reaches $\sim$32,300 (20,000–44,200), but that figure inherits the assumed $+4\%$ death continuation and is not an observation. The largest single uncertainty is not inside the simulation but in the choice of counterfactual, and it is not in the range above. Holding 2019 mortality flat is an assumption, and we fit the pre-crisis trend rather than assert it. The result is that **the trend is not distinguishable from zero on the available series**: the sparse 2006–2019 series (four years are unusable, see Disclaimers) gives $+0.236\%$/yr ($p = 0.066$, 95% CI -0.02 to $+0.49$) and the contiguous 2013–2019 window gives $+0.678\%$/yr ($p = 0.064$, CI -0.06 to $+1.41$). Both intervals contain zero. This is low power on seven to ten points, not non-identifiability in the formal sense. We therefore do *not* fit the trend and propagate it, which would manufacture precision the series cannot support; the flat anchor is a stated convention, not a finding. The two ends of the sensitivity below are not symmetric in status: the low end, 18,500 for 2025, is a bound Cuba's own data permit (the steeper window's upper limit, $+1.41\%$/yr); the high end, 31,600–38,000, is an *external* prior—the 0.5–1.5%/yr improvement Cuba's regional peers achieved—not a bound from the Cuban series, whose own opposite limit (-0.06%/yr) would give only $\sim$28,700. One further caveat belongs with these fits, and it is one-sided. The standardised series uses an open-ended 65+ group, so ageing *within* 65+ is uncontrolled and enters the slope as if it were rising mortality. Holding within-65+ rates fixed at their 2022 values, composition alone contributes $+0.160\%$/yr over 2013–2019 but -0.144%/yr over 2006–2019 (`scripts/composition_drift.py`, rates from Cuba's WHO submission over ONEI 3.3 denominators)—it changes sign between the two windows. The reason is a break in the denominator: ONEI

3.3's 85+ share of the 65+ population falls from 12.03% in 2012 to 11.15% in 2013, with the 85+ count dropping 9,168 while both younger elderly groups rise, which is not a demographic path. The contiguous window begins on that break and the sparse window straddles it, so the two are not measuring the same object, and that is a better explanation of the threefold slope difference than sampling noise. Because an inflated pre-crisis trend implies a higher counterfactual and therefore a smaller excess, this pushes the low end of the span down; we report the fits unadjusted and flag the direction. The choice of window is itself discretionary: the sparse window's upper bound (+0.49%/yr) gives ∼25,000 rather than 18,500. The full span, **18,500–38,000** (width 19,500), is of *comparable* width to the prior sensitivity range beside it (19,600) and nearly coincident with it—but the two are orthogonal and measure different things, so their near-identity is a coincidence a reader should not mistake for agreement. The counterfactual span is not propagated into the reported range. The same exercise on the cumulative window moves 2022–2025 to **45,200–103,300** from a deterministic flat-anchor value of 74,200 (the simulation median, quoted elsewhere and in the abstract, is 74,900; the 700 difference is the median-plug-in versus median-of-draws gap), so the headline is conditional on the flat anchor in the same way. Because the two are orthogonal, neither range alone is the total; we report no single combined interval, because there is no defensible prior over counterfactual choices to combine them with. The direction is genuinely ambiguous: Cuba's own fitted trends put the excess below the flat anchor (26,700 and 23,700), while the improvement its regional peers achieved would put it well above (31,600–38,000). We do not claim a direction. Turning to the anchor's internal noise rather than its choice: mean 2017–2019 age-specific rates sit within 1.1% of 2019 at ages 65+, where most deaths occur—slightly *above* it, meaning 2019 was a comparatively *good* year, so the anchor depresses expected deaths and mildly *inflates* the excess we report rather than flattering it downward—and the difference is well inside the 2% schedule noise.

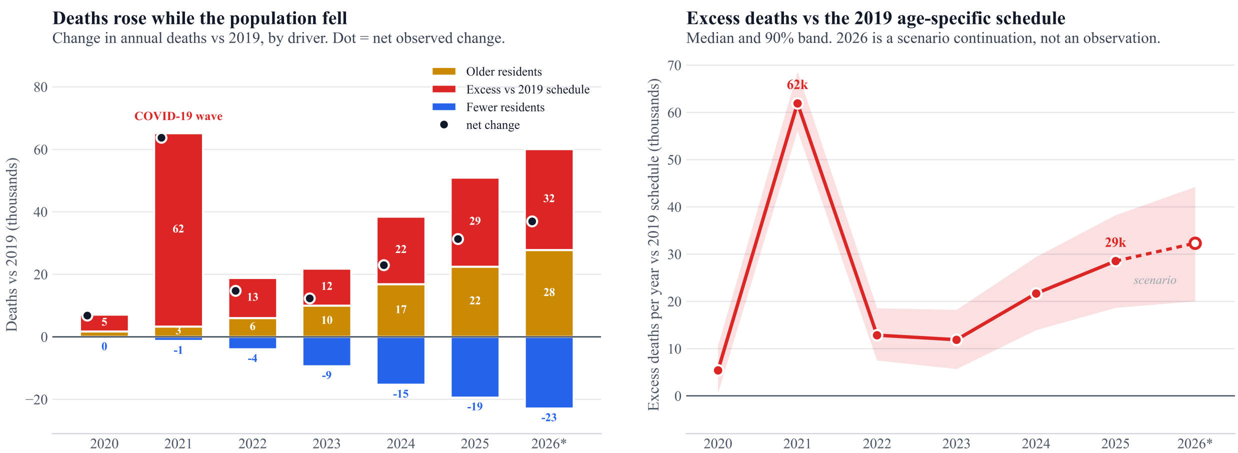


Figure 3: Why deaths rose while the population fell. Left: change in annual deaths versus 2019 decomposed into population size, ageing, and the change in age-specific rates; the dot is the net observed change including estimated unregistered deaths. Right: excess deaths against the 2019 schedule, median and prior sensitivity range. 2026 is a scenario continuation, not an observation. Bands are prior-predictive under the stated margins and share D_{fac} with the central scenario.

3.3 Projected population

Voter turnout fell sharply between 2018 and 2023; that pattern is compatible with physical absence but also with abstention, logistics, and political factors, and is *not* used as a person count.

Table 4 summarizes the three scenarios. The primary uncertainty statement is the across-scenario envelope: against the official end-2021 stock of 11.11 million, the gap by end-2025 spans **1.97–2.79 million**, or **17.8–25.1%**. The central scenario places living population at **8.58 million** (gap **2.53 million**, **22.8%**) against

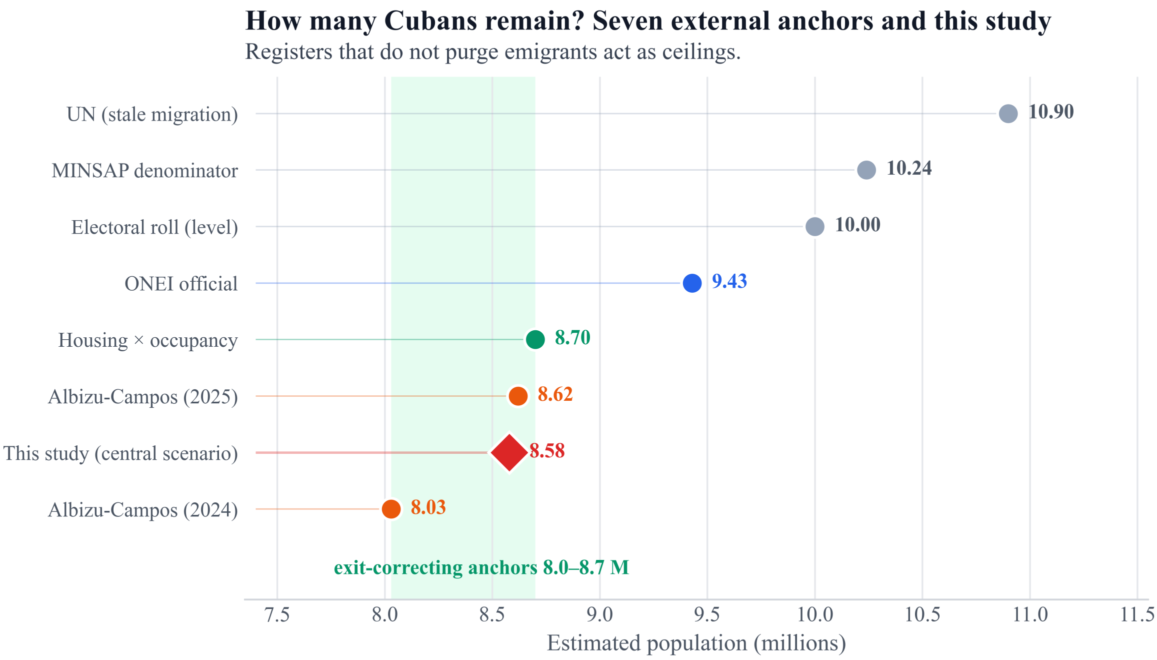


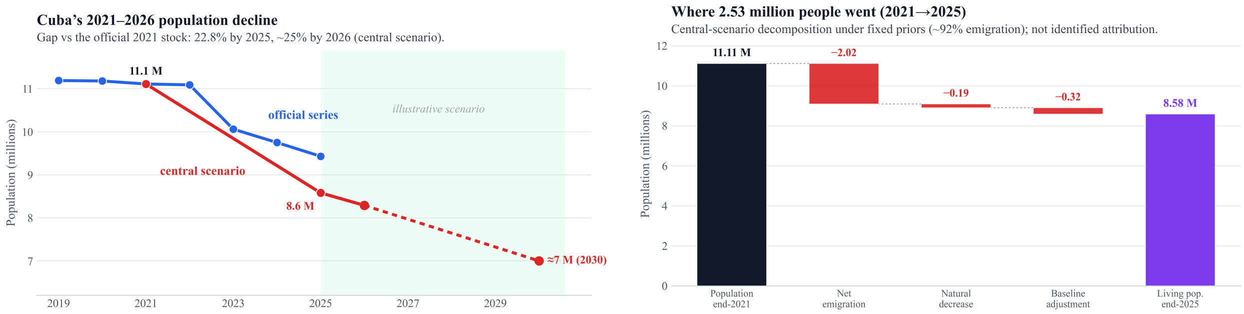
Figure 4: Triangulation of end-2025 population levels. the central scenario is the study estimand (not a validator); Albizu/housing are soft, assumption-laden anchors.

9.43 million officially, with the floor at 9.14 million and the upper stress case at 8.32 million. Emigration accounts for about **92%** of the central gap once the pre-2021 backlog is counted alongside the 2022–2025 flow; within the flow alone it is ∼91%. The end-2026 continuation places the central scenario near **8.29 million** (envelope 8.02–8.84 million). Because there is now a single base, the headline has a single form: between roughly one in six and one in four of the officially counted 2021 population is not there, with the central scenario near one in four – one in five.

Table 4: Three scenarios against one base: the official end-2021 stock (11,113,215). Point estimates are closed form; within-scenario 90% bands are prior sensitivity ranges over the stated margins. They are not confidence intervals, carry no coverage property, and no parameter in this paper is informed by a likelihood. Earlier versions carried five prior sets; a crisis-adjusted variant (within 0.02 M of the central path) and a vital-reconstruction companion have been retired.

Scenario	Gap (M)	Gap %	Pop. end-2025 (M)	Pop. end-2026 (M)
Null: ONEI at face value	*1.68*	*15.1*	*9.43*	*9.14*
Conservative	1.97	17.8	9.14	8.84
Crisis-adjusted central	**2.53**	**22.8**	**8.58**	**8.29**
Upper stress	2.79	25.1	8.32	8.02
Envelope (excl. null)	*1.97–2.79*	*17.8–25.1*	*8.32–9.14*	*8.02–8.84*

Regionally, Latin America and the Caribbean still grow (∼+0.7% per year; Table 6, UN WPP / CEPAL approximations). Cuba shows net population loss of about −3.2% annually on recent official-path comparisons (9,748,007 to 9,434,593). Among peacetime emigration-driven contractions, Venezuela is the closest regional peer (Table 5); disaster-driven cases are not treated as peers. By **end-2026**, the central continuation implies about 300,000 further loss: living population near **8.29 million** (across-scenario band 8.02–8.84 M). By end-2026 the upper scenario reaches a gap of ∼3.1 million; the central scenario ∼2.8 million (∼25%).



(a) Official versus scenario living population, 2021–2030 (the central scenario illustrative path in red; post-2026 segment is a non-probabilistic trend scenario).

(b) Accounting decomposition under the central scenario medians (2021–2025); illustrative scenario, not identified attribution.

Figure 5: Headline population path and accounting decomposition.

Table 5: Cuba versus selected peacetime emigration-driven contractions. Cuba row uses the central scenario against the official 2021 base and therefore includes a pre-2021 register correction (U_0) plus ONEI's own 2023 backlog catch-up; it is a stock discrepancy, not a 2021–25 flow, and is not like-for-like with the comparators, which are population losses. Comparators are approximate and not window-harmonised. Puerto Rico row confounded by hurricane disaster.

Case	Cum. loss	Approx. annual	Dominant engine
Venezuela 2013–24	∼25%	1–3%	emigration
Puerto Rico post-2017 (María era)	large net exit	—	emigration + disaster
Cuba 2021–25 (central)	∼22.8% (official base)	∼3–5%	emigration + internal

4 Discussion

4.1 Migration

Even after the 2023 revision, 2024–2025 net balances remain only slightly above U.S.-only recorded entries, while the 2025 balance admits 317,320 gross external emigrations (net −245,264): a shortfall against the multi-destination floor, though the 2025 balance is internally consistent. This paper therefore treats the official headcount as a ceiling and contrasts it with exit-correcting sources (Figure 4).

4.2 Excess mortality against the 2019 schedule

The one documented instance of the death register being revised downward is instructive. We use the published figure, but the pre-revision total would raise 2022 excess mortality from ∼12,800 to ∼21,800 and the 2022–2025 post-pandemic total from ∼74,900 to ∼83,800. This is the empirical anchor for how much slack the register demonstrably has: 7.5% in a single year, just below the 9% cap on D_{fac}, which is why that cap is retained rather than raised. The inference is weaker than it looks: a *downward* revision shows the register can shed deaths after publication, which bounds its revision volatility, not the number of deaths it is missing. Using that magnitude to cap an *under*-count multiplier is a heuristic, not an observation of under-count. It cuts both ways—a register that can shed 8,951 deaths without explanation is not fully trustworthy, but one that published a 54% single-year surge in 2021 (109,080 to 167,645) is also not one that hides at scale. The pre-revision figure is a press report [Barrenechea, 2023], not a figure we hold in an ONEI table, and is statused accordingly.

The two engines meet in the mortality decomposition. Emigration removes working-age adults, which raises the elderly share, which raises deaths even with an unchanged health system: that is the ∼22,400

Table 6: Annual population growth: Cuba versus the region (approximate, rounded). **Row provenance differs and the comparison is not like-for-like:** the peer rows are UN WPP 2024 multi-year averages [United Nations, Department of Economic and Social Affairs, Population Division, 2024], whereas Cuba's is a single-year change on ONEI's revised official series (9,748,007 to 9,434,593), which sits immediately after a register-revision seam. WPP itself holds Cuba near 10.9 M and would not produce this row. Read the Cuba figure as the official path's own recent pace, not as a WPP-comparable rate.

Country / region	Annual growth
Guatemala	$+1.7\%$
Bolivia	$+1.4\%$
Dominican Republic	$+1.0\%$
Mexico	$+0.8\%$
Colombia	$+0.7\%$
Latin America & Caribbean	$+0.7\%$
Brazil	$+0.4\%$
Chile	$+0.3\%$
Uruguay	$+0.1\%$
Venezuela	$\approx 0\%$ (after $\sim$25% loss 2013–24)
Cuba	-3.2%

additional deaths a year the ageing term measures. How much of that ageing is migration-induced is *not* identified here—the term is computed from the observed ONEI age structure, and Cuba's fertility transition was already generating ageing decades before the crisis—so it should be read as ageing from all causes combined, of which selective exit is one. On top of it sits a rate deterioration of $\sim$28,500 deaths a year that ageing does not explain, concentrated in 2024–2025 and still rising in the 2026 scenario. The policy-relevant asymmetry is that emigration can reverse, whereas neither those deaths nor the births that did not occur can. Two properties make the excess estimate more robust than the population scenarios it sits beside: it is far less exposed to the migration dispute than the crude bridge it replaces, because indirect standardisation leaves E_t moving only 3,070 deaths between the official and central-scenario stocks—though 3,070 is 11% of the 2025 headline, so it is not insulated, and the elderly count is *not* held fixed (see Methods); and it is insensitive to the anchor's *level*, which 2017–2019 sensitivity leaves essentially unchanged at ages 60+ (it is *not* insensitive to the anchor's trend, which is the dominant uncertainty above). It remains conditional on registered deaths being close to complete—an assumption the infant-mortality sentinel makes plausible for institutional deaths but does not establish, and which the D_{fac} prior is meant to absorb rather than resolve. Its principal weakness is the opposite of the population models': the mechanism is unidentified. Blackouts, fuel-driven ambulance and dialysis failures, drug shortages, delayed oncology, and the 2025–2026 arbovirus waves are all plausible pathways, and none is separately estimated here; the estimate is an association with the crisis period, not a cause-of-death attribution. WFP food-insecurity and anaemia indicators, and the fall in agricultural GDP, are discussed only as *contextual* amplifying pathways [World Food Programme, 2024]; they do not enter D_{fac} or the decomposition [Karlinsky and Kobak, 2021, Msemburi et al., 2023].

4.3 Projected population

The central scenario is retained because the IMR consistency check shrinks the need for large hidden-death multipliers and focuses residual uncertainty on migration [García et al., 2019, Pérez-Riverol, 2026b]. It is not an identified point estimate: absolute loss remains $\sim$91% migratory under the central scenario's priors, and U_0 with $M > 1$ can share a common outdated-register defect. The envelope leads, but not because it is wider: the central scenario's 90% band (0.90 M wide on the end-2025 stock) is in fact *wider* than the across-scenario envelope (0.82 M) and clears its lower end, though its upper end sits inside the envelope. The two measure different things—the band is a prior-predictive spread conditional on one assumption set, the envelope is the span across assumption sets—and the envelope leads because disagreement about assumptions, not sampling within them, is what actually drives the answer. The scientific lead is the 0.85 million by which the central scenario falls below ONEI's own end-2025 figure. The 2.53 million (22.8%) gap against the official 2021 base is the sum of that correction and the 1.68 million ONEI has itself already published; the two must not be quoted interchangeably. Exit is age- and sex-selective; it ages the island and structurally depresses fertility [Coleman, 2006, Díaz-Briquets, 2014, Lee and Mason, 2014]. The natural balance, positive in 2019, fell to −68,150 in 2025.

A 2026 census conducted under standard practice would be informative; the working end-2026 scenario band is 8.02–8.84 million rather than the 10.9 million World Population Prospects path under still-low net-emigration assumptions [United Nations, Department of Economic and Social Affairs, Population Division, 2024].

5 Disclaimers

Limitations. There has been no census since 2012; ONEI's ∼24-month emigration rule lags recent exits; destination microdata are incomplete; U.S. "settled" administrative totals are not ACS stock; housing-occupancy assumptions are fragile; ageing sketches are not cohort-component projections; life expectancy re-expresses mortality; 2026–2030 paths are scenarios. U_0 and M can double-count register lag; destination encounter ceilings can inflate unique persons; within-scenario intervals are conditional on one assumption set and should not be read as bounding the structural uncertainty, which the envelope carries instead. The mortality decomposition uses four age groups because that is the finest set ONEI 3.15 and 3.3 share; both ONEI 3.15 and 3.3 stop at 2022, so the 2023–2026 age structure is a cohort projection rather than an observation, and within-65+ composition after 2022 enters as a residual drift parameter. Its 2019 anchor assumes no counterfactual post-2019 improvement in Cuban mortality, and it shares D_{fac} with the central scenario, so it is not independent of the headcount scenario. Excess deaths against the 2019 schedule are an association with the crisis period, not a cause-of-death attribution. Food-insecurity indicators are contextual only.

The pre-crisis age-standardised series cannot be fitted on a complete window: ONEI 3.15 panels for 2009–2012 parse to roughly half the true death totals and are rejected by a reader guard, so the trend rests on 10 non-contiguous years (or 7 contiguous ones, which give a trend nearly three times larger). On either window the slope is statistically indistinguishable from zero, so no pre-crisis trend is available to fit, and the flat anchor should be read as a convention. The dominant uncertainty in the mortality estimand is the choice of counterfactual, not anything inside the simulation, and it is reported separately rather than folded into the range.

Status. This is a preprint and has not been peer reviewed. The three scenarios are prior narratives, not identified estimators of a unique living population, and reported bands are prior-predictive spreads rather than sampling error from data.

Data and code availability. ONEI tables and a machine-readable claim sheet are in the public repository `https://github.com/ypriverol/cubascience` under `demographics/`. Interactive companion: `https://ypriverol.github.io/cubascience/demographics/`. Monte Carlo scripts (`scripts/model_*.py`) and the mortality decomposition (`scripts/attributable_mortality.py`) write summaries to `artifacts/`. Supplementary PDF: `manuscript/en/cuba-depopulation-2021-2026-su` Riverol, 2026c]. Third-party literature is cited, not redistributed.

Competing interests. None declared.

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